URBAN EYE

GeoAI is the future

How GeoAI-based and Data-Driven Planning Drive the Urban Renewal Towards Success in ESG

Case Study in Ma Tau Kok

NG CHUN HIN | ISSAC

CURRENT PROBLEMS

Current Urban Development Process

Study	Plan	Implementation
Inception and Baseline Review	Formulation of Planning Framework; Evaluation and Updating; Public Engagement	Study Finalization
Manual Assessment	Manual Data Collection & Analysis	Manual Decision Making

Induced Problems

Environment

- Inadequate Insights on Prioritization
- Lack of District-based Considerations

Social

- Systemless (Urban Vibrancy)
- Unsustainable community planning

Governance

- Insufficient Public Engagement
- Time Consuming
- High Cost

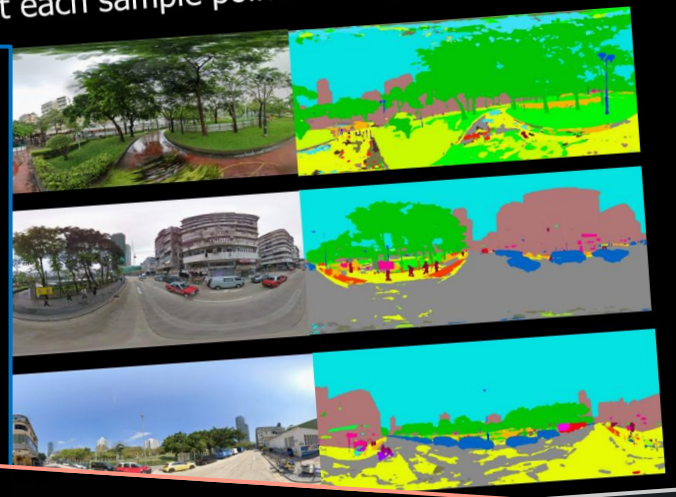
PROPOSED SOLUTIONS

13 Department / Bureau

24 Sets of Spatial Data

Semantic Segmentation

- Evaluation of the proportion of green views and sky views at each sample points through AI



Object Detection

- Images or videos undergo object detection to evaluate the vibrancy level at each sample points

Geographic Information System

Data Extraction
- 360-degree street view image

Data Analysis
- Spatial analytical techniques

Data Visualization
- Hexagonal grid maps

Natural Language Processing (NLP)

- Large language model (LLM) and OpenAI's gpt-3.5-turbo are used to develop an interactive virtual assistants that can carry out authentic and engaging conversations

LlamaIndex

OpenAI

Deep learning (Neural Network)

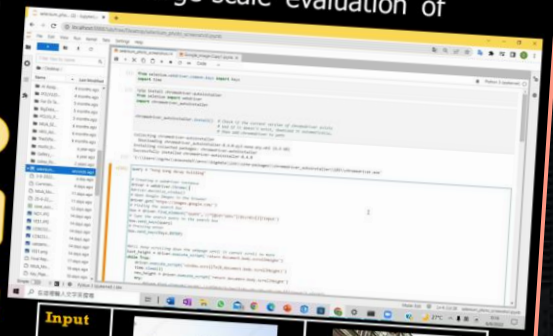
- A self-trained CV technique for large-scale evaluation of the building's physical conditions of external wall

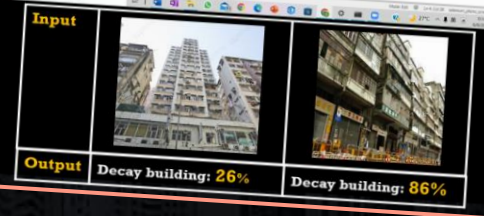
Qualified (n=3500)

Decay Buildings | Healthy Buildings

Training | Testing

Convolutional Neural Network

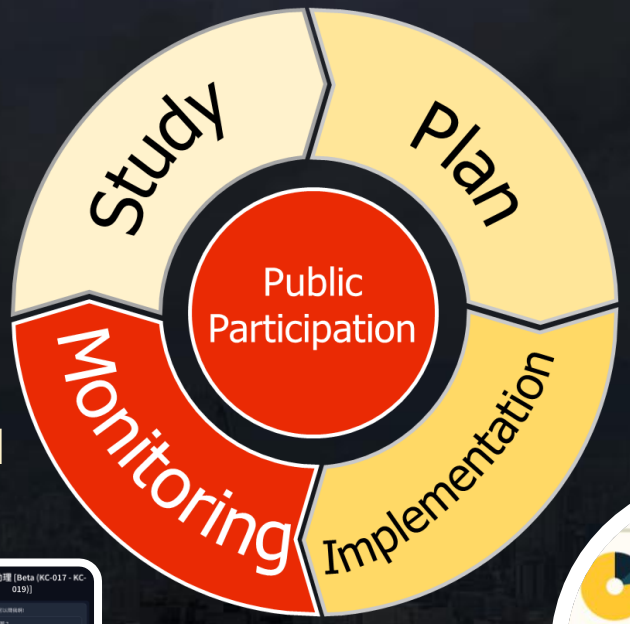




PROPOSED SOLUTIONS

1 URBAN VIBRANCY EVALUATION

Comprehend the Target Region



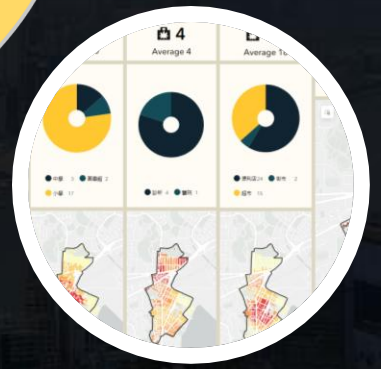
2 RENEWAL POTENTIAL SITE SEARCHING

Site Searching for Urban Renewal

4 PUBLIC-GOV COLLABORATION

Public Participation for Renewal

3 TEN-MINUTE COMMUNITY PLANNING



Planning for Identified Sites

By embracing data-driven and GeoAI-based strategies, an innovative idea is to transition from a linear working pattern to a cyclical approach for urban renewal. This transformative shift, supported by advanced GeoAI and spatial data analysis, are beneficial to continuous monitoring, adaptive interventions, and informed decision-making throughout the entire renewal process

ABOUT THE PROJECT

Relevant Stakeholders

Expected Values

R&D Stage

Organizations	Example
❑ Universities	香港大學建築學院 fia; Department of Building & Real Estate 建築及房地產學系
❑ NGOs	規劃情報; 本研
❑ NPOs	CSDI

Project Branding

BRAND BRAND BRAND BRAND BRAND

Project Profitability

$ $ $ $ $

E S G

To B

Organizations	Example
❑ Consultant Firm	ARUP; AECOM
❑ Developers	Hysan; 新世界發展有限公司 New World Development Company Limited

Project Branding

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Project Profitability

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Organizations	Example
❑ Urban Renewal Authority	市區重建局 Urban Renewal Authority
❑ Government B/D	規劃署 Planning Department; 運輸署 Transport Department; 地政總署 Lands Department

Project Branding

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Project Profitability

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